<table>
<tr><td>Lesson Code</td><td>2022-08-01</td><td>Level</td><td>Elementary</td><td>Total Time</td><td>40mins</td><td>Teaching Week</td><td colspan="2">Aug (01 – 05)</td></tr>
<tr><td>Subject</td><td>Mathematics</td><td>Books</td><td colspan="2">New countdown 7 2nd Edition</td><td>Page No.</td><td>1 – 9</td><td>Class</td><td>7</td></tr>
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Day	**Chapter Topic**	**Objectives:** **The students will be able to:**	**Teaching Resources:**	**Teaching Learning Strategies**
Day 1	**Unit 1: Operations on Sets**	• understand the basic concepts of sets. • revise the basic concepts of sets.	Textbook Worksheet	**Introduction;** To seek the prior knowledge of the students, ask the following question: such as • What is set? • Describe sets in tabular form. **Methodology**: • Write the sets below on the board. (a) The set of first five even numbers E = {2, 4, 6, 8, 10} • Ask the students which one is tabular form and which one is builder form. • If students will not give the right answer then define the tabular and set builder form. • Teacher will explain given worksheet. Students will complete the worksheet. **Wrap up:** Summarize the lesson by highlighting the key points. i.e. **Write the following sets in the set builder form.** (a) A = {2, 4, 6, 8} (b) B = {3, 9, 27, 81} (c) C = {1, 4, 9, 16, 25}
Day 2	**Unit 1: Operations on Sets**	• Define the different types of sets.	Textbook	**Introduction:** • Teacher will revise the sets impotence and previous lesson. **Methodology** • Teacher will revise the concept by explainingdifferent type of sets given on page 7 and 8. • Students will write all definition and examples in their notebooks. **Homework:** • Learn the definitions of different type of sets.
Day 3	**Unit 1: Operations on Sets**	• recognize the symbols used in sets. • identify equivalent and non-equivalent sets. • operateon objects to make non-equivalent sets equivalent. • comprehendthe concept of equivalent sets, union and intersection of sets	Textbook	**Introduction:** Recall the previous lesson ask following questions from students: • Differentiate between finite and infinite set. • Give an example of an equal set. **Methodology** Teacher will • define the cardinal number of sets. • Will explain the symbols used in sets given on pg8. • Discuss the concept of equivalent set, 1-1 correspondence • Tell the students that each element of set A has its correspondence of set B. These two sets has 1- 1 correspondence so these two sets are equivalent sets. • Also define the union of sets Students will solve Q3 & 4 of Exercise.1 in their notebooks. **Homework:** • Learn and re-practice the work done in class work.

Day 4	**Unit 1: Operations on Sets**	• define and understand the concept of intersection of sets and difference of sets	Textbook	**Introduction** • To recapitulate the previous lesson, following questions may be asked. e.g. • Define intersection of sets. • Define Difference of sets. **Methodology** • Introduce the concept of intersection of sets and difference of sets. • Explain Q5, from exercise 1. • Students will solve Q5 given in ex 1 in their notebooks.

Worksheet

Name: ________________________ Class: VII Date: ___________

Topic: Set Representation

1. Write the following sets in the roster form.

(a) A = The set of all even numbers less than 12

(b) B = The set of all prime numbers greater than 1 but less than 29

(c) C = The set of integers lying between -2 and 2

(d) D = The set of letters in the word LOYAL

(e) E = The set of vowels in the word CHOICE

(f) F = The set of all factors of 36

(g) G = {x : x ∈ N, 5 < x < 12}

(h) H = {x : x is a multiple of 3 and x < 21}

(i) I = {x : x is perfect cube 27 < x < 216}

(j) J = {x : x = 5n - 3,n ∈ W, and n < 3}

(k) M = {x : x is a positive integer and x2 < 40}

(l) N = {x : x is a positive integer and is a divisor of 18}

(m) P = {x : x is an integer and x + 1 = 1}

(n) Q = {x : x is a color in the rainbow}

2. Write each of the following in set builder form.

(a) A = {5, 10, 15, 20}

(b) B = {l, 2, 3, 6, 9, 18}

(c) C = {P, R, I, N, C, A, L}

(d) D = {0}

(e) E = { }

(f) F = {0, 1, 2, 3, ..., 19}

(g) G = {—8, —6, —4, —2}

(h) H = {Jan, June, July}

(i) I = {a, e, i, 0, u}

(j) J = {a, b, c, d, ……, z}

(k) K = {1/1, 1/2, 1/3, 1/4, 1/5, 1/6}

(l) L = {1, 3, 5, 7, 9}

Answer key

1. Write the following sets in the roster form.

(a) {2, 4, 6, 8, 10}
(b) {2, 3, 5, 7, 11, 13, 17, 19, 23}
(c) {-1, 0, 2}
(d) {L, O, Y, A}
(e) {O, I, E}
(f) {1, 2, 3, 4, 6, 9, 12, 18, 36}
(g) {6, 7, 8, 9, 10, 11}
(h) {3, 6, 9, 12, 15, 18}
(i) {64, 125}
(j) {-3, 2, 7}
(k) {1, 2, 3, 4, 5, 6}
(l) {1, 2, 3, 6, 9, 18}
(m) {O}
(n) {red, orange, yellow, green, blue, indigo, violet}

2. Write each of the following in set builder form.

(a) {x : x is a multiple of 5 and $5 \leq x \leq 20$}
(b) {x : x is a factor of 18}
(c) {x : x is a letter of the word 'Principal'}
(d) {x : $x \in W$ and $x < 1$}
(e) {x : $x \in N$ and $x < 1$)
(f) {x : $x \in W$ and $0 \leq x \leq 19$}
(g) {x : $x = -2n$ and $n \in N$ and $1 \leq n \leq 4$}
(h) {x : x is a month of the year beginning with J}
(i) {x : x is a vowel of the English alphabet}
(j) {x : x is a letter of the English alphabet}
(k) {x : $x = 1/x$, $n \in N$ and $1 \leq n \leq 6$}
(l) {x : x is odd, $x \leq 9$}